



STUDY PROTOCOL

Ethical gamified health communication intervention to prevent Work-related Musculoskeletal Disorders (WMSDs) in Myanmar migrants at Thailand's seafood factory: A study protocol

[version 1; peer review: awaiting peer review]

Phyu Hnin Hlaing ¹, Phaik Yeong Cheah ^{2,3}, Smith Boonchutima ^{1,4}

¹Faculty of Communication Arts, Chulalongkorn University, Bangkok, Thailand, 10330, Thailand

²Mahidol-Oxford Tropical Medicine Research Unit, Faculty of Tropical Medicine, Mahidol University, Bangkok, Bangkok, 10400, Thailand

³Centre for Tropical Medicine and Global Health, Nuffield Department of Medicine, University of Oxford, Oxford, England, UK

⁴Center of Excellence in Communication Innovation for Development of Quality of Life and Sustainability, Chulalongkorn University, Bangkok, Bangkok, 10330, Thailand

V1 First published: 28 Jun 2024, 9:347
<https://doi.org/10.12688/wellcomeopenres.21428.1>

Latest published: 28 Jun 2024, 9:347
<https://doi.org/10.12688/wellcomeopenres.21428.1>

Open Peer Review

Approval Status Awaiting Peer Review

Any reports and responses or comments on the article can be found at the end of the article.

Abstract

Background

Thailand's seafood industry heavily depends on Myanmar migrant workers, who face significant occupational health issues, including an increase in work-related Musculoskeletal Disorders (WMSDs). Due to linguistic and cultural limitations, traditional health communication strategies often fail to reach this population. This study explores the effectiveness of a culturally and linguistically tailored gamified health communication intervention aimed at preventing WMSDs in Myanmar migrant workers in a Thai seafood factory.

Methods

The intervention is developed and then implemented in two phases. The objective of the study is to develop, implement, and evaluate a gamified health intervention for Myanmar migrant workers that will prevent WMSDs. The intervention is customized to the workers' cultural background and work natures through focus group discussions, key informant interviews, and expert reviews. The intervention then consists of a set of 12 exercises designed to prevent

the WMSDs. The Intervention's impact is assessed through pre- and post-intervention evaluations measuring changes in knowledge, awareness, and self-reported musculoskeletal health.

Results

Since this is a study protocol, precise findings are not yet available. The target audience is expected to adopt safe work practices more frequently, have better physical health, and have a greater understanding of WMSDs.

Conclusions

This protocol describes a novel way to use a culturally-tailored and ethically appropriate gamified health communication intervention to address the health issues (such as WMSDs) experienced by Myanmar migrant workers in Thailand's seafood industry. This technique may serve as a model for comparable health promotion initiatives in other labour-intensive industries and migrant worker groups, promoting better awareness and prevention of WMSDs.

Registration

Thai Clinical Trials Registry (TCTR) (TCTR20240515003, Registration Date: 15 May 2024).

Plain Language Summary

Many workers from Myanmar migrate to Thailand's seafood industry in search of better career possibilities. Despite their important role, they deal with health issues, particularly musculoskeletal diseases (MSDs), which are injuries to the muscles, tendons, and joints brought on by their work-related conditioning.

Our project intends to develop a health program that prevents MSDs in these workers by using gamified (or game-like) features. Our aim is to promote health literacy by means of games and social media like Facebook.

We talked with a few workers and observed their job to gain insight into the difficulties they encounter, particularly about their well-being and working environment.

We created a set of 12 simple exercises that target the body area most impacted by their work nature, such as the neck and back, we conferred with health experts.

We created a Facebook intervention with exercises, games, and quizzes, offering an engaging way to learn about health prevention through rewards and points for sharing.

Our program addresses the health issues of seafood industry

workers, caused by repetitive, demanding work, by teaching them prevention methods on a familiar platform in a fun and accessible manner.

We hope to see an improvement in the workers' knowledge about health, a decrease in musculoskeletal problems, and an increase in healthy behaviours. This study could also help develop similar health programs for workers in other industries.

The purpose of this study is to provide a pleasant and educational approach for Myanmar migrant workers employed in Thailand's seafood industry to learn about reducing health risks related to their profession. Our goal is to increase the effectiveness and attraction of health education for them by utilizing social media and gamified components.

Keywords

Myanmar migrant workers, Work-related Musculoskeletal Disorders, Gamified intervention, Health communication, Ethical health communication, Thailand seafood industry



This article is included in the [Mahidol Oxford](#)

[Tropical Medicine Research Unit \(MORU\)](#)

gateway.

Corresponding author: Smith Boonchutima (Smith.b@chula.ac.th)

Author roles: **Hlaing PH:** Conceptualization, Data Curation, Formal Analysis, Investigation, Methodology, Project Administration, Writing – Original Draft Preparation; **Cheah PY:** Conceptualization, Funding Acquisition, Supervision, Writing – Review & Editing; **Boonchutima S:** Conceptualization, Funding Acquisition, Methodology, Project Administration, Resources, Supervision, Writing – Review & Editing

Competing interests: No competing interests were disclosed.

Grant information: This work was supported by Wellcome [222870]; the Century Two Fund (C2F) at Chulalongkorn University, with Dr. Phyu Hnin Hlaing, the principal investigator and first author of this protocol, receiving the grant for her postdoctoral research project. Additional partial funding came from the Research Unit of the Faculty of Communication Arts at Chulalongkorn University. *The funders had no role in study design, data collection and analysis, decision to publish, or preparation of the manuscript.*

Copyright: © 2024 Hlaing PH *et al.* This is an open access article distributed under the terms of the [Creative Commons Attribution License](#), which permits unrestricted use, distribution, and reproduction in any medium, provided the original work is properly cited.

How to cite this article: Hlaing PH, Cheah PY and Boonchutima S. **Ethical gamified health communication intervention to prevent Work-related Musculoskeletal Disorders (WMSDs) in Myanmar migrants at Thailand's seafood factory: A study protocol [version 1; peer review: awaiting peer review]** Wellcome Open Research 2024, 9:347 <https://doi.org/10.12688/wellcomeopenres.21428.1>

First published: 28 Jun 2024, 9:347 <https://doi.org/10.12688/wellcomeopenres.21428.1>

Introduction

In the dynamic landscape of Thailand's seafood industry, a key demographic stands out: the Myanmar migrant workers, numbering around 334,000 across various districts with significant populations in places like Mae Sot, Mueang Ranong, and Mae Sai, have become indispensable to the industry's growth. Drawn to Thailand by the promise of better economic opportunities, these workers now represent a vital component of the workforce, especially noteworthy given that approximately 63,000 of them arrived post the military takeover in Myanmar. However, this migration brings forth pressing occupational health challenges, notably the escalating incidence of Work-related Musculoskeletal Disorders (WMSDs). [Koichi, 2010; Patel, 2023; Tran, 2016].

While prior studies have spotlighted the prevalence of WMSDs - conditions affecting muscles, tendons, and joints, caused or exacerbated by workplace activities and ergonomic stressors. Soe *et al.* (2015) emphasized the need for enhanced occupational safety measures to protect these workers¹. A study focusing on Cambodian farm workers found a significant correlation between the duration of work and the risk of developing neck and lower back pain, indicating the cumulative nature of WMSDs².

A closer examination of specific departments in frozen food manufacturing factories reveals various work-related stressors, including repetitive tasks, uncomfortable working postures³, and poor workplace ergonomics, all of which can lead to musculoskeletal issues affecting different body parts like the neck, lower back, and limbs⁴⁻⁸. These occupational health problems highlight the urgent need for strategic interventions that consider the unique cultural and linguistic characteristics of the Myanmar migrant worker community.

The prevention of WMSDs among workers, especially in high-demand settings like seafood factories, has garnered attention, focusing on various interventions that are effective in mitigating the risks associated with WMSDs. Notable interventions include modifications of equipment, adjustable workstation elements, scheduled work breaks, and stress management programs, which have been demonstrated to reduce WMSD outcomes such as pain and work ability⁹. The spectrum of interventions extends to workstation modification, changes in work methods, administrative controls, training, organized exercise, protective equipment usage, and medical management aimed at reducing exposure to musculoskeletal disorder risks¹⁰.

Migrant workers face challenges in seeking healthcare due to language barriers, legal status issues, and financial constraints¹¹. These challenges underscore the need for ethical considerations when creating health communication strategies tailored for this demographic. Given the vulnerable status of the migrant worker population, it's imperative to approach this matter with ethical mindfulness, ensuring cultural respect, informed consent, confidentiality, and unwavering commitment to participants' well-being and dignity. This ethical commitment lays

a strong foundation for the effective implementation of the proposed multilingual health communication intervention, thereby contributing towards creating a safer and more health-conscious working environment in the Thai food industry.

Integrating these interventions with social media gamification could provide a novel approach, particularly beneficial for migrant workers in seafood factories. Gamification, which involves the application of game elements (e.g., rewards, competitions) in non-game contexts, can serve as a cost-effective alternative to assist in the implementation of these interventions¹². Specifically in the context of musculoskeletal rehabilitation, gamification has been employed to increase motivation and adherence to rehabilitation protocols¹³. Social media platforms offer a conducive environment for gamification, which can lead to positive changes in physical activity levels and diet-related behaviours among individuals¹⁴. Utilizing social media gamification can potentially foster a more engaging and interactive environment for migrant workers to adhere to workplace interventions designed to prevent or mitigate WMSDs¹⁵⁻¹⁷. This integration could be particularly appealing to migrant workers, who might find the gamified, social, and interactive elements culturally engaging and motivating, thereby promoting better adherence to preventive measures and interventions. Furthermore, the digital nature of gamified interventions offers flexibility and accessibility, addressing the constraints faced by busy workers, especially migrant workers dealing with language barriers and limited access to traditional healthcare services, as they can access information during their free time.

Before the commencement of our study, our team undertook two vital preliminary activities to better understand the context and conditions faced by Myanmar migrant workers in the seafood factory. We obtained formal permission to conduct our study at a Frozen Food factory following IRB approval to ensure full compliance with organizational and ethical guidelines. Firstly, we conducted informal informational interviews with senior Myanmar workers at the Frozen Food factory in Samut Sakhon Province. The discussions shed light on their daily challenges, working conditions, and preferences for potential interventions. Secondly, an observational visit was organized at the Frozen Food factory, where our team closely observed the working conditions, tasks, and ergonomic aspects of the workers' roles on the production line. These preliminary activities were crucial in providing a foundational understanding of the real-life challenges faced by the workers and will inform the design and tailoring of our research interventions.

With the surge in digital engagement and the interactive nature of games, this study protocol explores into the viability of a gamified health communication intervention for Myanmar migrant workers in Thailand's seafood factories. By harnessing the appeal and interactivity of gamification, the study aims to bridge communication gaps, provide accessible health information, and foster an environment that prioritizes the prevention of WMSDs.

Protocol

Research objectives, questions and rationale

- Research Objectives:
 - To develop a culturally-tailored and ethically appropriate gamified health communication intervention to prevent WMSDs among Myanmar migrant workers in the seafood factory.
 - To validate the adapted questionnaire (hereafter referred to as the assessment form) for assessing knowledge and awareness of WMSDs and. Physical function among the target population.
 - To assess the effectiveness of a gamified health communication intervention, delivered through a Facebook Group, in preventing and reducing the incidence of WMSDs among Myanmar migrant workers in Thailand's seafood factory.
 - To recommend ergonomic improvements in the workplace environment to optimize workstations and practices, reducing the occurrence of WMSDs among Myanmar migrant workers.
- Research Questions
 - How can a culturally-tailored and ethically appropriate gamified health communication intervention be developed to effectively prevent Work-Related Musculoskeletal Disorders (WMSDs) among Myanmar migrant workers in the Seafood Factory?
 - How effective is the adapted questionnaire in accurately assessing the knowledge and awareness of work-related musculoskeletal disorders (WMSDs) and physical function among Myanmar migrant workers?
 - Following the implementation of the gamified health communication intervention, what are the observed changes in the knowledge and awareness levels regarding WMSDs, musculoskeletal function, and progression in healthy habits among the Myanmar migrant workers?
 - What specific ergonomic improvements can be recommended in the workplace environment to optimize workstations and practices, thereby reducing the occurrence of WMSDs among Myanmar migrant workers?
- Research Rationale
 - Why is this research important?

The well-being of Myanmar workers in Thailand is important in itself, and critical to the region's overall socio-economic structure as well as to the individual workers. As the introduction emphasized, a considerable percentage of these workers are impacted by a current concern known as Work-related Musculoskeletal Disorders (WMSDs). Addressing this issue affects not only the health of the individual but also the productivity of the workplace, the

economy, and the general standard of living for these people and their families.

Moreover, traditional interventions, while effective to some extent, might not be sustainable or engaging in the long run. There's a pressing need for innovative, scalable, and sustainable results that can be easily adopted and maintained by the workers. This research, by exploring the intersection of technology, social media, and health interventions, seeks to fill this gap.

- How can gamification and social media, specifically Facebook, potentially address this issue?

Gamification, or the use of game-design elements in non-game contexts, has shown potential for increasing motivation and engagement in a variety of fields, including business and education. Gamification is a promising strategy for health interventions since motivated and engaged people are more likely to take up and stick with new activities.

Facebook provides a special benefit as a platform. Due to its widespread use, this instrument is recognizable and easily accessed by workers in Myanmar. The barrier to entry is substantially lower because the intervention is delivered through a platform that they are already familiar with, and since almost everyone has a mobile phone, accessing Facebook is convenient. Likewise, Facebook's features, such as groups, polls, live videos, and reactions, provide a rich toolkit for creating an interactive and community-driven intervention.

Combining the principles of gamification with the features of Facebook can lead to a potent intervention strategy. For instance:

- Community Building: Workers can share their progress, challenges, and successes, fostering a sense of community and peer support.
- Instant Feedback: Through likes, comments, and reactions, workers receive immediate feedback, reinforcing positive behaviours.
- Milestones & Rewards: Gamified milestones can be set, and workers achieving them can be recognized, further motivating participation.

Methodology

Phase I- Intervention development

- Data Collection
 - **Focus Group Discussion (FGD)**

In our research, we plan to conduct Focus Group Discussions (FGDs) with approximately 30 Myanmar migrant workers, divided into groups of 8-10 until on data saturation. The participants will be categorized into three distinct groups for analysis: the 'Newcomer' group, consisting of individuals with 0-6 months of work experience, 'Mid-level

Worker' for 6 months -2 years experiences and the 'Senior Worker' group, comprising those with more than 2 years of experiences. The selection criteria for participants include Myanmar migrant workers aged 18–60 who are proficient in Burmese, while management staff, non-migrant workers, and individuals unwilling or unable to consent will be excluded.

Dr PHH, a Myanmar researcher, will lead the FGDs, with a Burmese doctor (Research Assistant) serving as a note-taker. The discussions, expected to last between 40-60 minutes, will take place in a compound near the staff dormitory. For the purpose of accuracy and thorough documentation, we will audio-record the sessions, ensuring confidentiality of the data. Our approach to participant recruitment involves initial contact through staff in charge, followed by information sessions to explain the FGD's purpose, process, risks, and benefits, and the distribution of Consent Forms. Participants will be compensated for their time with a combination of refreshment (one snack, one water bottle and one juice valued at 60 THB) and cash amount of 100 THB. The total value of the compensation will not exceed 200 THB, and participants will not incur any costs for their participation. The language used throughout the FGDs and all related materials will be Burmese/Myanmar, ensuring clear and effective communication with the participants.

○ **Key Informant Interview (KII)**

The Key Informant Interviews (KIIs) for our study will be led by two principal researchers: Dr PHH and Dr SB, both are qualified in related Musculoskeletal Disorders (WMSDs) and occupational health. The interviews will be conducted in Thai or Burmese, depending on the interviewee's preference. Dr. PHH will lead the interviews with Myanmar participants, while Dr Smith will interview the Thai participants. Each KII is planned to last 40-60 minutes in a private and comfortable setting within the factory premises, either in lobby or the cafeteria, based on the interviewee's preference. We will schedule these during off-peak hours to ensure a quiet and undisturbed environment. Audio recordings will be made with participant consent, facilitated by a research assistant handling notes and logistics.

Participants must meet the following inclusion criteria:

- who is involve in wellbeing of the migrant workers
 - who has authority in making decision on intervention to promote the migrant workers health
 - who has experiences more than 5 years
 - who evaluate the quality of work and manage the shifts.
- Those lacking this experience or unwilling or unable to provide informed consent will be excluded from the study. The tentative number of participants from each category are as follows:
- Thai HR personal (1-2 persons)
 - Burmese Line Managers (2-3 persons)
 - Thai Line Managers (2-3 persons)
- Participants will be approached through direct outreach and given an information sheet and consent form. For their contribution, interviewees will receive a souvenir (e.g. water bottle or mug) valued at no more than 200 THB.

○ **Experts Review**

The role of experts in this study is twofold. Approximately 3-5 experts, selected based on their professional background as physical or exercises physiologist with relevant experience, will participate in the Expert Reviews. The primary responsibilities are:

- **Reviewing the Gamified Health Communication Intervention:** Experts will assess the intervention's content, focusing on its relevance, accuracy, and potential to raise WMSD awareness among Myanmar workers in Thailand's seafood industry.

Integration with the Intervention Development Process: The expert review process starts with a presentation of the study's objectives and preliminary content, followed by discussions on the workers' job nature and postures. This exchange informs the experts' recommendations on appropriate preventative exercises and shapes the development of a detailed exercise prescription.

Gamified Health Communication Strategy: The intervention incorporates gamification elements like Points Rewards Systems, Quizzes, Weekly Competitions, Leaderboard Rewards, and Educational Posts. Preliminary titles, objectives, and expected outcomes have been crafted for these elements to ensure they are structured and goal-oriented. The experts' review will critically assess these to guarantee their educational effectiveness and relevance to the target audience.

- **Validating the Assessment Form:** The experts will also be responsible for validating the assessment form that will be used to evaluate the intervention's impact on the workers' knowledge, Attitudes, and practices regarding

WMSD prevention. The Assessment Form (AF) will be used in our study is divided into two parts: pre-intervention and post-intervention. The pre-intervention AF is comprised of four distinct sections:

1. Demographic information, which gathers basic participant details. No names, addresses or other identifiable details will be collected.
2. Knowledge and awareness regarding WMSDs through 10 carefully curated questions. Drawing on insights from existing research, this section aims to capture participants' understanding of key factors related to WMSDs.
3. Musculoskeletal problems assessment, this section is adapted from the Nordic Musculoskeletal questionnaire. This section employs a series of concise yes/no questions for each specific body part, facilitating a targeted evaluation of musculoskeletal problems. By incorporating established assessment tools, we enhance the reliability and validity of our data collection.
4. Initial awareness and attitudes towards workplace health, encompassing five specific questions.

The post-intervention AF, on the other hand, consists of five sections. The first three sections mirror those of the pre-intervention AF. The fourth section shifts focus to 'progress in health habits', with questions conceptually similar to the fourth section of the pre-intervention AF, aiming to assess changes in health behaviour and attitudes. The final and fifth section of the post-intervention AF is dedicated to 'Participant Reflections and Feedback on the intervention, providing valuable insights into the participants' perspectives on the effectiveness and impact of the intervention.

Adaptation of the Nordic Questionnaire (Part 3 of the AF): The Nordic Musculoskeletal Questionnaire (NMQ) is originally developed by Kuorinka *et al.* in 1987, the NMQ is renowned for its effectiveness in identifying the prevalence and impact of musculoskeletal symptoms in various occupational settings (Kuorinka, I., Jonsson, B., Kilbom, A., *et al.*, 1987, "Standardised Nordic questionnaires for the analysis of musculoskeletal symptoms". *Applied Ergonomics*, 18(3), 233–237). We have specifically adapted the NMQ for part 3 of our Assessment Form. While the NMQ traditionally presents questions for each body part separately, we have modified this approach for efficiency and clarity. The adaptation involves using the question style of the NMQ but consolidating into a single table that encompasses eight key body parts: Neck, Shoulders Elbows, Wrists/Hands, Upper Back, Lower Back, Hips/Thighs, Knees, Ankles/Feet.

The approach to identifying and recruiting experts, primarily physical and occupational therapists, is based on professional connections within the field. Most of these experts are Myanmar therapists, who are researcher's professional acquaintances. The initial contact with the experts is made informally via email or social media. This informal outreach is to gauge their interest and availability for participating in the study.

Upon receiving a positive response indicating their interest and availability, the researcher will send them an official email containing the Consent Form. Upon consent, a one-on-one call is scheduled to discuss their insights on WMSD awareness and the intervention content. Following this, experts are given an assessment evaluation form to review and feedback within a week. The informal outreach process is already initiated informally with some experts to ensure a smooth start to the Expert Review phase once IRB approval is granted.

Future Steps: Incorporating expert feedback is vital to developing an effective health communication intervention. The experts' roles in reviewing the gamified intervention content and validating the assessment form are central to creating a scientifically sound and engaging program. After the expert review, the intervention will proceed with finalizing the content and producing educational videos, which will be distributed through the Facebook group. This comprehensive approach aims to equip the migrant workers with the knowledge and skills necessary to prevent WMSD in their workplace.

- Data Collection Tools

- **Focus Group Discussion Guide:** is designed to facilitate discussions with Myanmar workers, aiming to develop culturally-tailored and ethical gamified health communication intervention to prevent WMSDs.
- **Key Informant Interview Guide:** is designed for collecting detailed insights from senior figures in Thailand's seafood industry, focusing on health communication, workplace culture, and the challenges faced by Myanmar migrant workers regarding WMSDs.
- **Assessment Form Validation Scale:** is structured tool designed to gather expert feedback on the relevance, clarity, cultural appropriateness, and overall effectiveness of an AF.

- Data Analyses

- **Thematic Analysis:** For qualitative data obtained from Focus Group Discussion and Key Informant Interviews we will use thematic analysis to identify patterns and themes.

- **Descriptive Statistics:** we will employ *Content Validity Index (CVI)* to summarize the quantitative feedback or ratings provided by the experts.

- **Content Validity Index (CVI)**

- **Expert Rating:** A panel of experts' rate item on "Assessment Validation Scale" using 1-5 scale (1= Strongly Disagree, 5= Strongly Agree)
- **Item Level CVI (I-CVI):** For each item, I-CVI will be calculated by determining the proportion of experts who rated the item as 4 (Agree) or 5 (Strongly Agree). The formula is $I-CVI = \frac{\text{Number of experts giving a rating of 4 or 5}}{\text{Total number of experts}}$.

- **Scale-Level CVI (S-CVI)**

- **S-CVI/Average (S-CVI/Ave):** We will find the average of the I-CVIs for all items. The formula is: $\text{Sum of I-CVIs} / \text{Number of items}$.

- **S-CVI/Universal Agreement (S-CVI/UA):** Calculate the proportion of items that all experts rated as 4 or 5.

- **Interpretation**

- An I-CVI close to 1 indicates strong content validity for an item.
- For S-CVI/Ave, a value of 0.90 or above is considered indicative of excellent content validity for the scale.
- For S-CVI/UA, a higher proportion indicates stronger agreement among experts on the scale's content validity.

The study flow chart is shown in [Figure 1](#).

- Ethical considerations

- **Informed Consent:** We will obtain informed consent from all participants, ensuring they fully understand the study's purpose, procedures, and any potential risks involved. During the recruitment process,

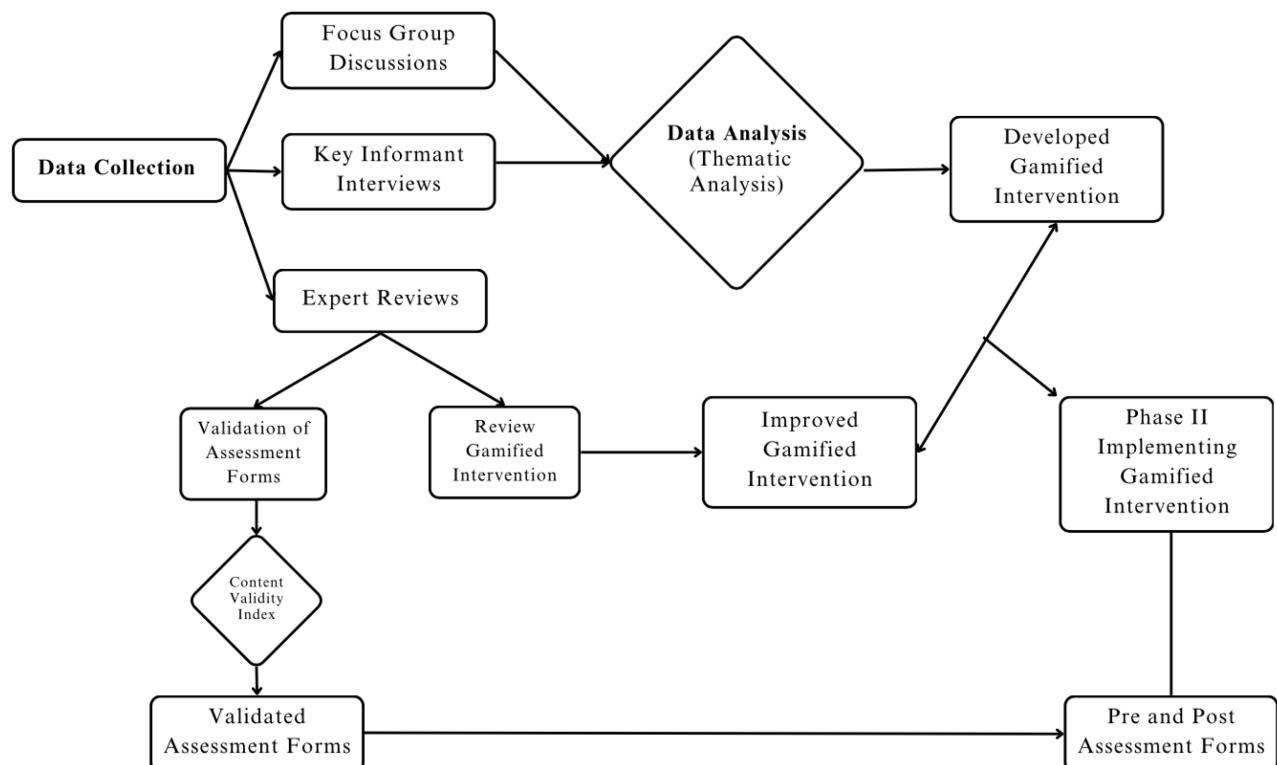


Figure 1. Flowchart depicting the process of developing a gamified intervention through two phases. The initial phase, Data Collection to develop the intervention, involves gathering information through Focus Group Discussions and Key Informant Interviews. The collected data is then subjected to thematic analysis in the Data Analysis phase, which informs the development of the initial gamified intervention. This intervention is then validated by experts, leading to a review of the initial design. Improvements are incorporated, resulting in an Improved Gamified Intervention. This intervention is further refined and undergoes a Phase II Implementation, developing in a finalized gamified intervention. The efficacy and impact of the intervention are measured using Pre and Post Assessment Forms, ensuring a comprehensive evaluation of the intervention's outcomes. The authors confirm that they own and have created this figure specifically for this submission.

prospective participants will receive information about the study from employers or union representatives or through researchers' network. They will sign a consent form after receiving a thorough explanation.

- **Confidentiality and Privacy:** We guarantee that participant information is kept private and confidential by securely storing it and anonymizing it before analysis. We will ensure participants that the information provided will not be revealed to their employers.
- **Safety and Well-being:** We will make sure that participant' pre-existing musculoskeletal disorders and other health concerns related to their jobs are not made worse by the study procedures, particularly the focus group discussion and observation.
- **Expert Review Transparency:** To preserve the study's ethical standing and credibility, we will be transparent in our selection of experts for the review process and demand a clear declaration of any potential conflicts of interest.

Phase II- Implementing Intervention

- Objectives
 - To implement the gamified health communication intervention among Myanmar migrant workers in the seafood factory.
 - To evaluate the intervention's impact on knowledge, awareness, and preventive behaviours concerning WMSDs using the validated assessment form.
- Participants

We plan to enrol 400 participants, factoring in a 10% dropout rate, to achieve an estimated final sample size of around 360 individuals, with this number derived from power analysis calculations to ensure statistical robustness and representativeness of the study population.

 - Inclusion Criteria
 - Myanmar workers aged 18 and above.
 - Active Facebook users (defined as those who log in at least once a week).
 - Willingness to participate in the entire duration of the intervention.
 - Written informed consent provided
 - Exclusion Criteria
 - Workers with diagnosed chronic musculoskeletal/ medical conditions that might interfere with the intervention.

- Workers planning to leave their job or the region within the next two months.

Recruitment Strategy

- **Direct Outreach:** we will approach workers through their employers or union representatives, introducing the research and its potential benefits.
- **Word of Mouth:** we will encourage early participants to spread the word among their peers.

Intervention Design for Preventative Health Communication

Objective and Rationale

The intervention design aims to address musculoskeletal disorders (MSDs) among migrant workers in the Frozen Factory, specifically targeting those engaged in long hours of shrimp peeling and other repetitive tasks. The design is informed by the observed risk of upper¹⁸ and lower crossed syndromes¹⁹ among these workers, conditions that stem from their prolonged standing and forward-leaning postures. These syndromes result in muscular imbalances: a combination of weakened and tight muscles in the neck, shoulders, back, and hip areas.

Exercise Selection and Sequence

The intervention comprises 12 exercises, carefully chosen and sequenced from easiest to more challenging, to ensure gradual engagement and effectiveness. The sequence starts with exercises that address the most immediate discomforts and progresses towards those that strengthen and support long-term musculoskeletal health:

1. **Chin Tucks:** Targets neck muscles with minimal effort, addressing neck strain and promoting good posture.
2. **Neck Tilt Stretch:** Eases neck and shoulder tension, simple to execute from a seated or standing position.
3. **Child's Pose:** Stretches the back and is gentle enough for beginners, offering a broad range of relaxation for the back muscles.
4. **Palm Stretch:** Simple and effective for relieving wrist and hand tightness, relevant for workers performing repetitive hand tasks.
5. **Hip Flexor Stretch:** Addresses lower body posture and is relatively easy to perform, reducing anterior pelvic tilt.
6. **Chest Stretch (Pectoral Stretch):** Reduces chest tightness and is crucial for counteracting forward hunching postures.
7. **Thumb Touches:** Improves thumb mobility and reduces stiffness, relevant for manual dexterity.

8. **Tendon Gliding Exercises:** Enhances finger and hand tendon mobility, preventing locking and stiffness.
9. **Sponge Squeeze:** Increases hand strength and promotes tendon flexibility, suitable for improving grip and manual tasks.
10. **Glute Bridge:** Strengthens the posterior chain, a bit more demanding but essential for pelvic and lower back stability.
11. **Plank:** Builds core strength and improves overall posture, requiring more strength and endurance.
12. **Prone Cobra/Seated Row:** More challenging for upper back strength but vital for shoulder alignment and back support.

This progression is designed to first alleviate tension and improve mobility in areas most affected by the workers' repetitive tasks, gradually building up to strengthening exercises that can be performed with minimal equipment and space, considering the factory environment and the workers' living conditions.

○ Implementation and Accessibility

The exercise videos, demonstrating each activity, will be shared through the Intervention Facebook group, offering easily accessible, detailed instructions. The program prioritizes exercises that require the least amount of space and equipment possible, taking into account the living and working conditions of manufacturing workers.

○ Justification for the Exercise Selection

The selected exercises specifically target the muscle groups affected by the workers' daily tasks. Focused prevention is required because of the predominance of upper crossed syndrome, which is characterized by forward head posture and rounded shoulders²⁰, and lower crossed syndrome, which is indicated by anterior pelvic tilt. These conditions are common in workers who stand for long periods and engage in repetitive motions, like shrimp peeling, leading to specific muscular weaknesses and tightness that the exercises aim to address²¹.

○ Monitoring and Feedback

Regular feedback will be solicited from the workers to gauge the exercises' effectiveness and their impact on musculoskeletal health. This feedback will be crucial for refining the intervention, ensuring it remains relevant and beneficial for the target population.

• Content Delivery

○ Informational Posts

Three times a week, the Facebook group will feature posts on topics crucial to WMSDs prevention, such as

posture care, ergonomic practices, exercise training, and stretching exercises. These posts aim to educate workers and provide them with practical knowledge to prevent WMSDs.

○ Quizzes

To reinforce learning and engage participants, quizzes related to the informational posts will be shared on alternate days. This pacing allows time for participants to absorb the information before testing their understanding.

○ Weekly Challenges

Each week will feature a challenge that prompts participants to demonstrate a preventive measure or their comprehension of a specific topic. These challenges are designed to be applicable in real-world settings, promoting the practical application of acquired knowledge.

• Gamification Features

○ Points System

Points are provided for different activities, so participants can earn points for their participation. For example, they will receive 1 point for like a post, 2 points for commenting, 3 points for intelligent comments, 1 point for completing a quiz, 2 points for answering the quiz correctly, and 5 points for taking part in weekly challenges.

○ Leaderboard

The top 10% of active participants will be shown on a leaderboard, encouraging competition and recognizing participation. The leaderboard, which is expected to include 400 participants, will be divided into three categories: the top 10, players ranked 11–20, and participants ranked 21–50.

The gamified intervention is shown in [Figure 2](#).

• Data Collection

○ **Pre-Intervention Assessment:** Developed Assessment Form will be used to assess participants' knowledge and awareness of work-related musculoskeletal disorders (WMSDs) and to assess musculoskeletal problems. This will help establish a baseline.

○ **Post-Intervention Assessment:** Unlike the pre-intervention assessment, the questionnaire, which now includes five additional questions to assess participants' stage of behaviour change toward healthy habits, will be utilized to evaluate the effectiveness of the intervention.

○ **Engagement Tracking:** We will monitor participants' engagement levels by tracking their reactions, comments, and posts within the Facebook group.

Gamified Intervention

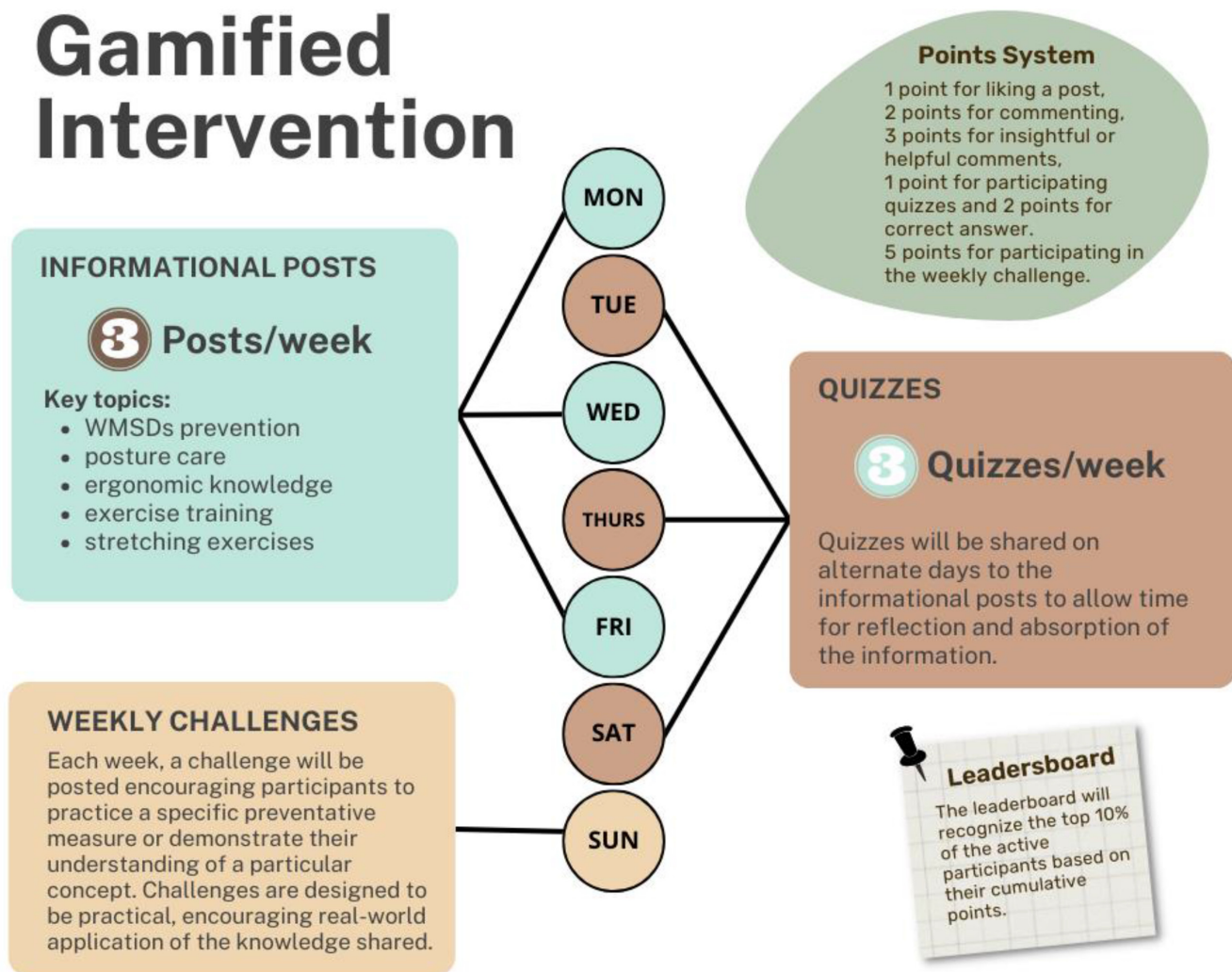


Figure 2. Schematic overview of a gamified intervention program structured over a week. The intervention includes three main components: Informational Posts, Quizzes, and Weekly Challenges. Informational Posts are shared three times a week, covering topics such as WMDs prevention, posture care, ergonomic knowledge, and stretching exercises. Quizzes, occurring thrice a week on alternating days, facilitate knowledge retention and reflection. Weekly Challenges prompt active engagement with the material, requiring participants to demonstrate understanding or application of concepts in a practical manner. A points system rewards engagement and learning, with points allocated for participation, quiz performance, and challenge participation, culminating in a Leaderboard that tracks the top participants. The schematic serves as a guide for the structure and rewards system of the intervention. The authors confirm that they own and have created this figure specifically for this submission.

- This can be done manually or by using analytics tools integrated with Facebook.
- Adherence Monitoring: We will track the consistency of participants in performing daily challenges and participating in weekly competitions.
- Data Analysis
 - Statistical Methods: We will use descriptive statistics to summarize the data. Employ inferential statistics, such as t-tests or ANOVA, to compare pre and post-intervention data and determine the intervention's effectiveness.
- Engagement Analysis: Engagement metrics (likes, comments, shares) will be analysed to determine the level of active participation and its correlation with the intervention's outcomes.
- Ethical Considerations
 - **Informed Consent:**
 - **Participant Information and Consent Forms:** Each participant will get a participant information sheet and consent form outlining their rights as participants, any risks, and the objective of the research before they participate.

The study will be explained in full to potential participants throughout the recruitment process, after which they will sign a consent form and get an information sheet from employers or union representatives.

- **Clarification Sessions:** Prior to providing consent, participants can ask questions regarding the research and get any uncertainties answered during our Facebook Live sessions or Zoom meetings.
- **Ongoing Consent:** We will consistently remind participants that they are under no obligation to continue participating in the study at any time.
- **Privacy and Confidentiality:**
 - **Private Facebook Group:** The intervention will be conducted in a private Facebook group where only approved members can view and participate in the activities. This ensures that outsiders cannot access the participants' data or interactions related to the study.
 - **Data Handling:** Any data extracted from Facebook for analysis will be anonymized. Personal identifiers will be removed or coded to ensure that individual participants cannot be identified in any reports or publications.
 - **Data Storage:** All digital data will be stored on encrypted drives or secure cloud storage platforms with restricted access.
- **Potential Risks:**
 - **Emotional Distress:** Discussing musculoskeletal pain might bring up feelings of frustration or distress for some participants. To mitigate this, provide resources or contacts for counselling or support services.
 - **Data Breach:** While using Facebook, there's always a risk of data breaches. While we cannot control Facebook's security, we will ensure that our data handling and storage practices are secure and compliant with best practices.
 - **Misinformation:** We will ensure that all information and exercises shared are evaluated by experts to prevent the spread of misinformation or potentially harmful advice.
- **Benefits:**
 - The exercises and behaviours participants engage in during the intervention may result in a decrease in musculoskeletal symptoms.
 - Participants will acquire knowledge regarding musculoskeletal health, which they can utilize

in their everyday lives and share with their peers.

- Participating in a community that offers support can yield emotional and motivational advantages for its members.
- The participation of participants will enhance the scientific understanding of WMSD prevention and the possibility of gamified therapies, which will be advantageous for future researchers and workers in the field.

Conclusion

The integration of a gamified health communication intervention within Thailand's seafood industry, specifically targeting Myanmar migrant workers, addresses the urgent need to mitigate Work-related Musculoskeletal Disorders (WMSDs). The carefully selected and sequenced 12 exercises form the core of this intervention, tailored to counteract the prevalent upper and lower crossed syndromes caused by long hours of repetitive tasks such as shrimp peeling. This intervention's design, based on thorough initial research and expert validation, underscores its potential to effectively enhance musculoskeletal health and overall well-being among this vulnerable worker population.

The use of a Facebook-based platform for delivering this intervention leverages the digital engagement trends and the familiarity of social media among the target demographic, ensuring accessibility and sustained participation. The gamified elements, integrated with educational content and practical exercises, are anticipated to foster an engaging, interactive, and supportive community environment conducive to learning and behaviour change.

Discussion

This intervention development is grounded in a deep understanding of the distinct occupational and sociocultural backgrounds of Myanmar migrant workers in Thailand's seafood sector. Prior studies, such as those by Soe *et al.* (2015)¹, have highlighted the prevalence of musculoskeletal disorders among Myanmar migrants, underscoring the urgency of addressing these health concerns. The informal interviews and observational visits conducted in the initial stages of our research provided valuable insights into the workers' unique demands and obstacles, echoing findings from Patel and Ghosh (2023)⁸ that ergonomic stressors in seafood processing significantly contribute to WMSDs.

The use of gamification in health communication, as suggested by Warsinsky *et al.* (2021)⁶, is an innovative approach that holds significant potential for influencing WMSD preventive strategies, especially in multicultural and linguistically varied work settings. This intervention is designed to enhance knowledge, modify behaviour, and reduce the frequency and severity of WMSDs among the workers, fostering active participation and offering immediate, material rewards for engagement. The gamification strategy draws on the principles discussed

by Chen (2014)¹², who emphasizes the role of digital game-play in facilitating social inclusion and health education among migrant workers.

The expected success and positive outcomes of this intervention could serve as a model for similar programs in other industries and areas with high migrant worker populations. This aligns with the findings of Cotton and Patel (2019)¹⁷, who observed the effectiveness of gamified approaches in promoting health and fitness in diverse populations. Furthermore, it opens avenues for further research into the efficacy of gamified health interventions across different occupational settings, as indicated by the systematic reviews and meta-analyses conducted by Tersa-Miralles *et al.* (2022)⁹.

In summary, this protocol offers a systematic, evidence-based strategy to tackle the critical issue of WMSDs among Myanmar migrant workers in Thailand's seafood industry. Integrating state-of-the-art gamification techniques with traditional health communication methods and leveraging the widespread use of social media, this intervention is poised to create a sustainable and impactful health enhancement program.

Ethics and consent

This study was conducted in strict accordance with ethical standards. The research involving human participants was reviewed and approved by the Research Ethics Review Committee for Research Involving Human Research Participants, Group I, Chulalongkorn University, Bangkok, Thailand. The committee approved this study under Certificate of Approval No. 052/67. The approval was granted on February 28, 2024, and is valid until February 27, 2025. The study title associated with this approval is "Gamified Health Communication Intervention for Preventing Work-Related Musculoskeletal Disorders (WMSDs) among Myanmar Migrant Workers in Thailand's Seafood Factory."

All participants will provide informed consent, and all methods will be carried out in accordance with relevant guidelines and regulations, including the Belmont Report (1979), the Declaration of Helsinki (2013).

Data availability

Underlying data

No data are associated with this article.

Extended data

Open Science Framework: Gamified Health Communication Intervention. <https://doi.org/10.17605/OSF.IO/WDG3C22>

This project contains the following underlying data:

- Focus Group Discussion Guides (English and Myanmar Versions)
- Key Informant Interview Guides (English and Myanmar Versions)
- Pre-Intervention Assessment Forms (English and Myanmar Versions)
- Post-Intervention Assessment Forms (English and Myanmar Versions)
- Assessment Form Validation Rating Scale
- Participant Information Sheet and Consent Forms (for all groups of participants)
- Reporting Guidelines (COREQ checklist and Gamification Reporting Guideline)
- Completed SPIRIT checklist

Data are available under the terms of the [Creative Commons Attribution 4.0 International license](#) (CC-BY 4.0).

Acknowledgements

We would like to express our gratitude to the Frozen Food Co. Ltd (the study site) for their collaboration and important help in making this research possible. We clarify that the company had no role in the study design, data collection and analysis, decision to publish, or preparation of the manuscript, and they do not have any financial or non-financial interests that could have influenced the results of this study. We also express our gratitude to the Myanmar migrant labourers, whose participation and insights were crucial to our study. We appreciate the guidance and expertise that the interdisciplinary team and our colleagues have provided us; their collaboration has been essential to the success of this project.

References

1. Soe KT, Laosee O, Limsatchapanich S, *et al.*: **Prevalence and risk factors of musculoskeletal disorders among Myanmar migrant workers in Thai seafood industries.** *Int J Occup Saf Ergon.* 2015; **21**(4): 539–46. [PubMed Abstract](#) | [Publisher Full Text](#)
2. Thetkathuek A, Meepadit P, Sa-ngiamsak T: **A cross-sectional study of musculoskeletal symptoms and risk factors in Cambodian fruit farm workers in Eastern Region, Thailand.** *Saf Health Work.* 2018; **9**(2): 192–202. [PubMed Abstract](#) | [Publisher Full Text](#) | [Free Full Text](#)
3. Govaerts R, Tassignon B, Ghillebert J, *et al.*: **Prevalence and incidence of work-related musculoskeletal disorders in secondary industries of 21st century**

- Europe: a systematic review and meta-analysis.** *BMC Musculoskelet Disord.* 2021; **22**(1): 751.
[PubMed Abstract](#) | [Publisher Full Text](#) | [Free Full Text](#)
4. Kyaw KWW, Platt L, Biji M, *et al.*: **The effect of different types of migration on symptoms of anxiety or depression and experience of violence among people who use or inject drugs in Kachin State, Myanmar.** *Harm Reduct J.* 2023; **20**(1): 45.
[PubMed Abstract](#) | [Publisher Full Text](#) | [Free Full Text](#)
 5. Thetkathuek A, Meepradit P, Jaidee W: **Factors affecting the musculoskeletal disorders of workers in the frozen food manufacturing factories in Thailand.** *Int J Occup Saf Ergon.* 2016; **22**(1): 49–56.
[PubMed Abstract](#) | [Publisher Full Text](#)
 6. Warsinsky S, Schmidt-Kraepelin M, Rank S, *et al.*: **Conceptual ambiguity surrounding gamification and serious games in health care: literature review and development of game-based intervention reporting guidelines (GAMING).** *J Med Internet Res.* 2021; **23**(9): e30390.
[PubMed Abstract](#) | [Publisher Full Text](#) | [Free Full Text](#)
 7. Aasmoe L, Bang B, Egeness C, *et al.*: **Musculoskeletal symptoms among seafood production workers in North Norway.** *Occup Med (Lond).* 2008; **58**(1): 64–70.
[PubMed Abstract](#) | [Publisher Full Text](#)
 8. Patel J, Ghosh T: **An ergonomic evaluation of the prevalence of musculoskeletal disorders among fish processing workers of Suri.** *Biomed.* 2023; **43**(1): 21–25.
[Reference Source](#)
 9. Tera-Miralles C, Bravo C, Bellon F, *et al.*: **Effectiveness of workplace exercise interventions in the treatment of musculoskeletal disorders in office workers: a systematic review.** *BMJ Open.* 2022; **12**(1): e054288.
[PubMed Abstract](#) | [Publisher Full Text](#) | [Free Full Text](#)
 10. Base NRC (US) SC for the W on W-RMITR: **Intervention to control musculoskeletal disorders.** 1999.
[Reference Source](#)
 11. Aung TNN, Shirayama Y, Moolphate S, *et al.*: **Health risk behaviors, musculoskeletal disorders and associated cultural adaptation, depression: a survey among Myanmar migrant workers in Chiangmai, Northern Thailand.** *Int J Gen Med.* 2019; **12**: 283–92.
[PubMed Abstract](#) | [Publisher Full Text](#) | [Free Full Text](#)
 12. Chen VHH: **Facilitating social inclusion of migrant workers through digital game play.** *GlobDev* 2014. 2014.
[Reference Source](#)
 13. Proffitt R: **Gamification in rehabilitation: finding the “just-right-challenge.”** 2024; 132–57.
[Publisher Full Text](#)
 14. Goodyear VA, Wood G, Skinner B, *et al.*: **The effect of social media interventions on physical activity and dietary behaviours in young people and adults: a systematic review.** *Int J Behav Nutr Phys Act.* 2021; **18**(1): 72.
[PubMed Abstract](#) | [Publisher Full Text](#) | [Free Full Text](#)
 15. Goodyear VA, Boardley I, Chiou SY, *et al.*: **Social media use informing behaviours related to physical activity, diet and quality of life during COVID-19: a mixed methods study.** *BMC Public Health.* 2021; **21**(1): 1333.
[PubMed Abstract](#) | [Publisher Full Text](#) | [Free Full Text](#)
 16. Bamidis PD, Gabarron E, Hors-Fraile S, *et al.*: **Gamification and behavioral change: techniques for health social media.** *Particip Heal through Soc Media.* 2016; 112–35.
[Publisher Full Text](#)
 17. Cotton V, Patel MS: **Gamification use and design in popular health and fitness mobile applications.** *Am J Health Promot.* 2019; **33**(3): 448–51.
[PubMed Abstract](#) | [Publisher Full Text](#) | [Free Full Text](#)
 18. **Upper-crossed syndrome - physiopedia.**
[Reference Source](#)
 19. **Lower crossed syndrome - physiopedia.**
[Reference Source](#)
 20. Chang MC, Choo YJ, Hong K, *et al.*: **Treatment of upper crossed syndrome: a narrative systematic review.** *Healthcare (Basel).* 2023; **11**(16): 2328.
[PubMed Abstract](#) | [Publisher Full Text](#) | [Free Full Text](#)
 21. Bayattork M, Seidi F, Minoonejad H, *et al.*: **The effectiveness of a comprehensive corrective exercises program and subsequent detraining on alignment, muscle activation, and movement pattern in men with upper crossed syndrome: protocol for a parallel-group randomized controlled trial.** *Trials.* 2020; **21**(1): 255.
[PubMed Abstract](#) | [Publisher Full Text](#) | [Free Full Text](#)
 22. Hlaing PH: **Gamified health communication intervention.** *OSF.* [Dataset]. 2024.
<http://www.doi.org/10.17605/OSF.IO/WDG3C>